

Introduction to RStudio

Day 1 - Introduction to Data Analysis with R

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October 7, 2026

Difference between R and RStudio



Analogy and image from [ModernDive Book](#)

R is the **programming language** and the **program** that does the actual work

RStudio is the **integrated development environment (IDE)**.

Interface to R, syntax highlighting, file management, ...

A quick tour around RStudio

The screenshot displays the RStudio IDE interface. The main editor window shows a script named 'penguin_script.R' with the following R code:

```
1 # Script to plot some data
2 library(ggplot2)
3 library(palmerpenguins)
4
5 head(penguins)
6
7 ggplot(penguins, aes(x = flipper_length_mm, y = body_mass_g, color = species)) +
8   geom_point() +
9   geom_smooth(method = "lm") +
10  scale_color_manual(values = c("darkorange", "purple", "cyan4")) +
11  theme_bw()
12
```

The Environment pane on the right shows the 'Global Environment' with a dataset 'someData' containing 10 observations of 2 variables. The 'Values' pane shows the first three rows of 'someData':

variableA	variableB
num [1:3]	1 2 3
	10.5

The Console pane at the bottom shows the output of the R script, including the R license notice, version information, and the execution of the script:

```
R 4.0.3 · C:/Users/Selina_User/Files_Selina/Repos/02_workshops/intro-to-r/
R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

Warning messages:
1: package 'devtools' was built under R version 4.0.5
2: package 'usethis' was built under R version 4.0.5
3: package 'reprex' was built under R version 4.0.5
> variableA <- c(1,2,3)
> variableB <- 10.5
> someData <- data.frame(a=1:10, b=1:10)
> 2+2
[1] 4
> print("hello")
[1] "hello"
>
```

The File Explorer pane on the right shows the directory structure of the project, including files like '.gitignore', 'day_00', 'day_01', 'day_02', 'day_03', 'day_04', 'example_files', 'example.html', 'example.Rmd', 'intro-to-r.Rproj', and 'libs'.

A quick tour around RStudio

The screenshot shows the RStudio interface with four panels highlighted by blue boxes and labels:

- Scripts:** The top-left panel shows a script named `penguin_script.R` with the following code:

```
1 # Script to plot some data
2 library(ggplot2)
3 library(palmerpenguins)
4
5 head(penguins)
6
7 ggplot(penguins, aes(x = flipper_length_mm, y = body_mass_g, color = species)) +
8   geom_point() +
9   geom_smooth(method = "lm") +
10  scale_color_manual(values = c("darkorange", "purple", "cyan4")) +
11  theme_bw()
12
```
- Environment:** The top-right panel shows the Environment pane with a table of variables:

variableA	num [1:3]	1	2	3
variableB				
- Console:** The bottom-left panel shows the R console output:

```
R 4.0.3 - C:/Users/Selina_User/Files_Selina/Repos/02_workshops/intro-to-r/
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> variableA <- c(1,2,3)
> variableB <- 10.5
> someData <- data.frame(a=1:10, b=1:10)
> 2+2
[1] 4
> print("hello")
[1] "hello"
>
```
- Files, Plots & more:** The bottom-right panel shows the Files pane with a list of files and folders in the directory `C:/Users/Selina_User/Files_Selina/Repos/02_workshops/intro-to-r`. The files are:

Name	Size	Modified
..		
.gitignore	113 B	Jun 16, 2021, 11:18 PM
day_00		
day_01		
day_02		
day_03		
day_04		
example_files		
example.html	29.6 KB	Jun 16, 2021, 10:18 PM
example.Rmd	17.6 KB	Jun 16, 2021, 9:57 PM
intro-to-r.Rproj	293 B	Jun 17, 2021, 1:45 PM
libs		

Console pane

- Execute R code
- Output from R code in scripts is printed there
- Type a command into the console and execute with **Enter/Return**



Tip

Use arrow keys to bring back last commands

```
R version 4.0.3 (2020-10-10) -- "Bunny-Wunnies Freak Out"
Copyright (C) 2020 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64 (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

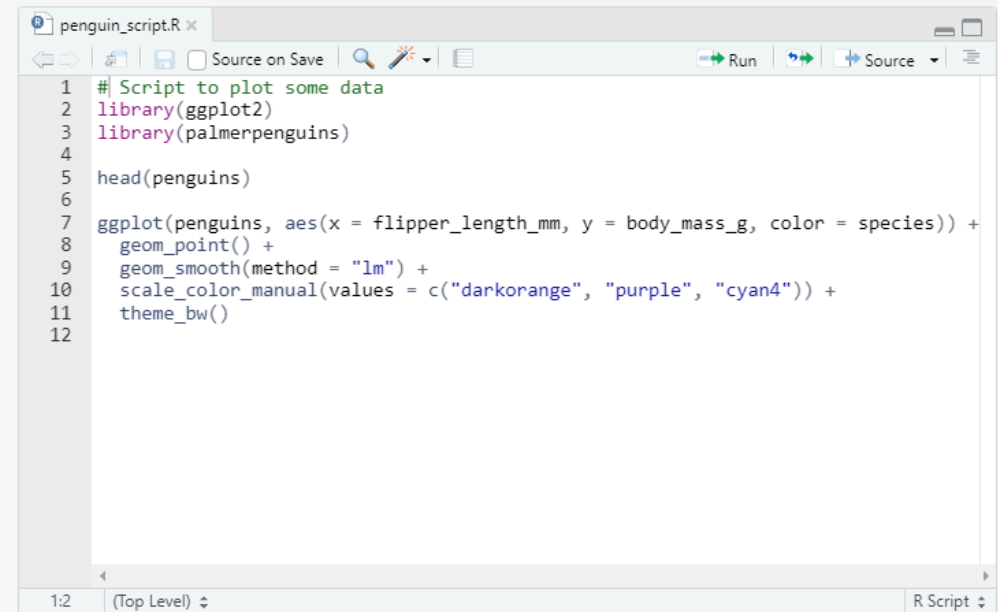
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Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
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Warning messages:
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2: package 'usethis' was built under R version 4.0.5
3: package 'reprex' was built under R version 4.0.5
> 4+4
[1] 8
> print("hello")
[1] "hello"
> |
```

Script pane

- Write scripts with R code
 - Scripts are text files with R commands (file ending `.R`)
 - Use scripts to save commands for reuse



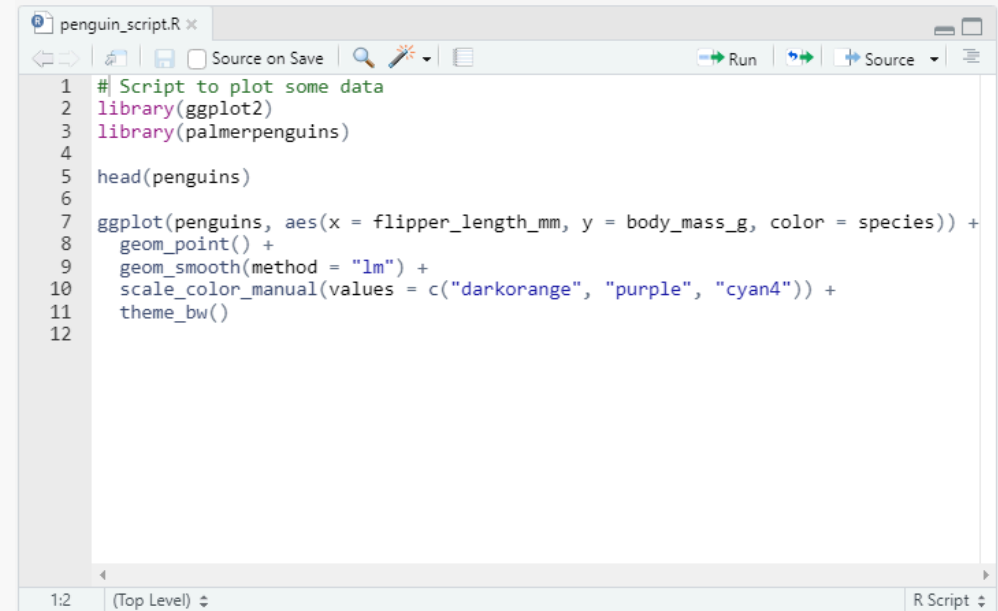
```
penguin_script.R x
Source on Save
Run
Source

1 # Script to plot some data
2 library(ggplot2)
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5 head(penguins)
6
7 ggplot(penguins, aes(x = flipper_length_mm, y = body_mass_g, color = species)) +
8   geom_point() +
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10  scale_color_manual(values = c("darkorange", "purple", "cyan4")) +
11  theme_bw()
12
```

1:2 (Top Level) ↕ R Script ↕

Script pane

- Create a new R script:
File -> New File -> R Script
- Save an R script:
File->Save (Ctrl/Cmd + S)
- Run code line by line with **Run button**
(Ctrl+Enter/Cmd+Return)
- You can open multiple scripts



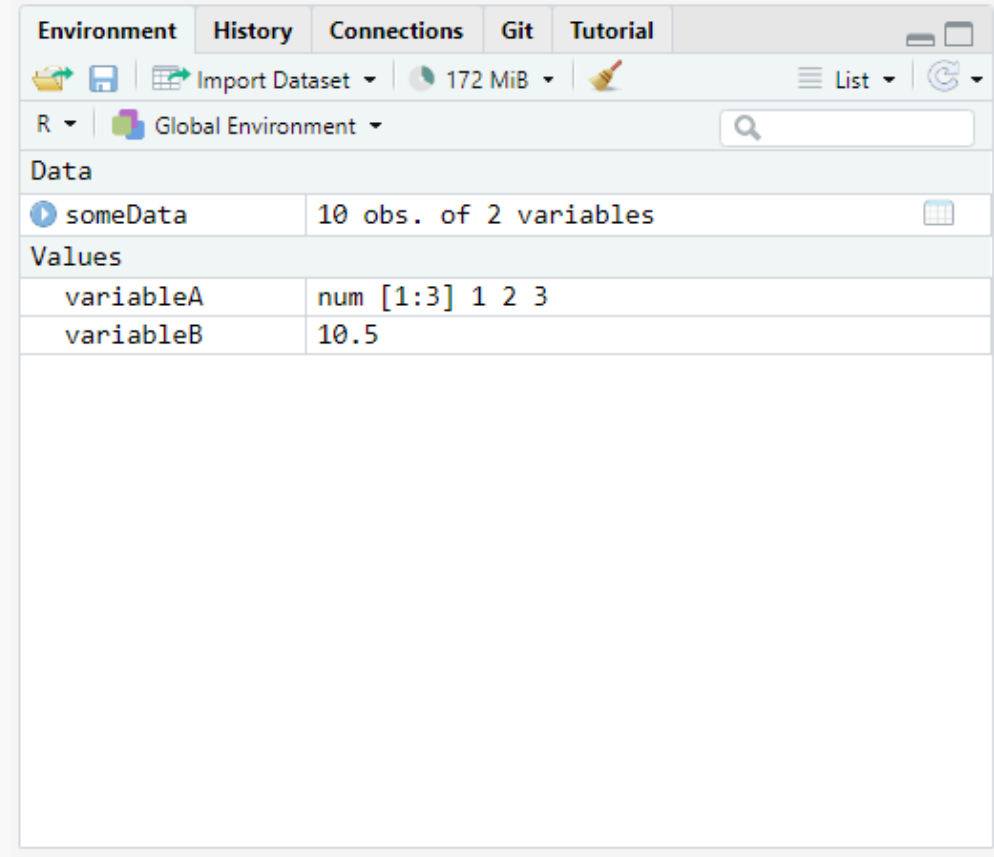
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11  theme_bw()
12
```

💡 Script vs Console

Use **scripts** for all your analysis and for commands that you want to save.
Use **console** for temporary commands, e.g. to test something.

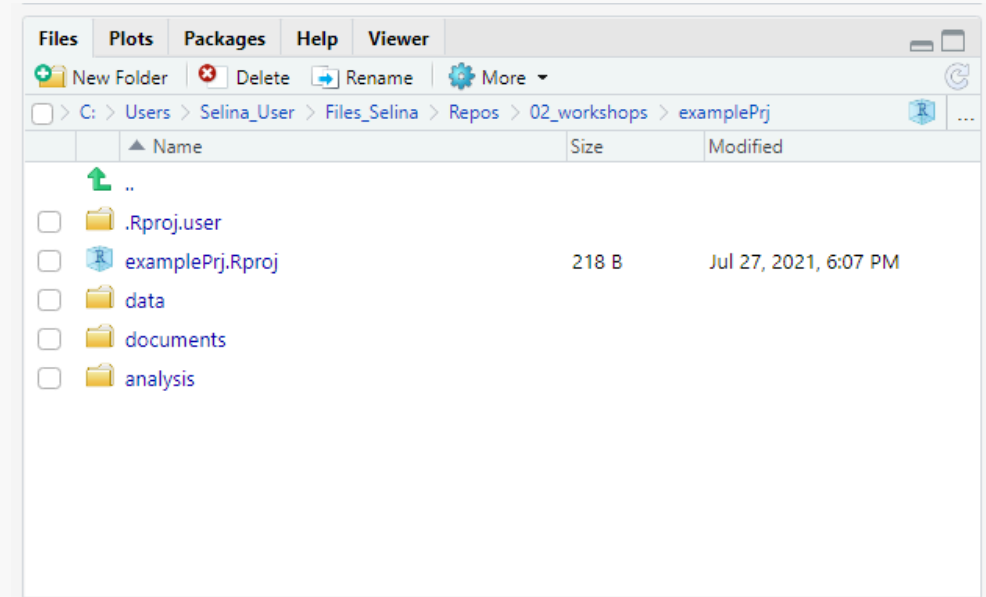
Environment pane

- Shows objects currently present in the R session
- Is empty if you start R



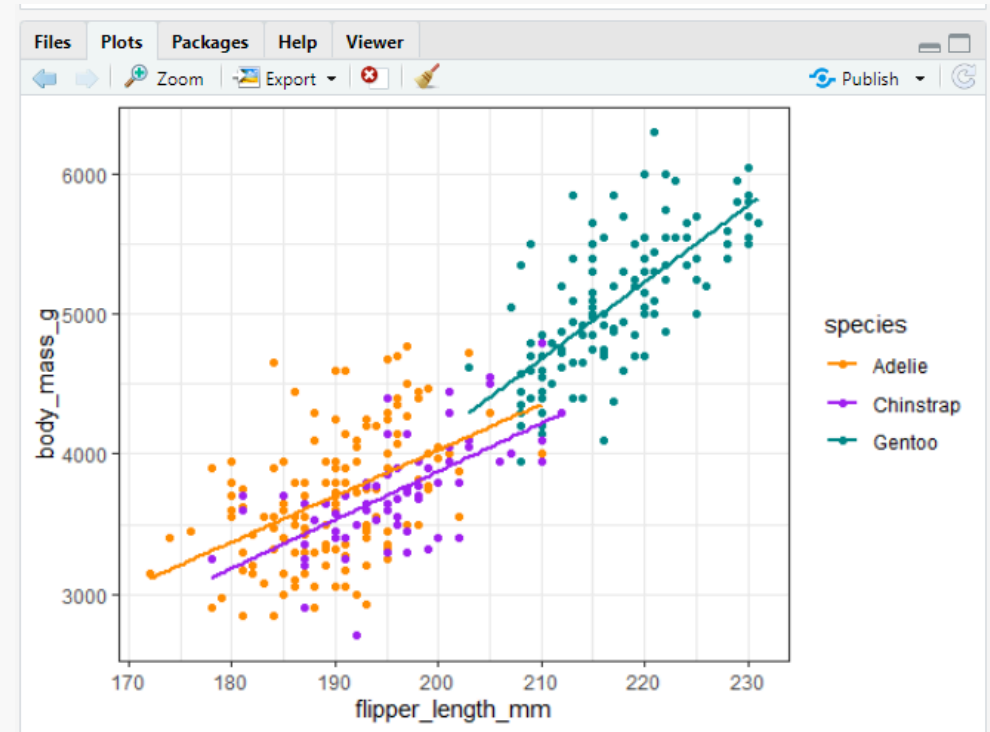
Files pane

- Similar to Explorer/Finder
- Browse project structure and files
 - Find and open files
 - Create new folders
 - Delete and rename files
 - ...



Plot pane

- Plots that are created with R will be shown here



Project-oriented workflow

One project = one folder

- Contains everything: data, scripts, figures, ...
- Nothing important is somewhere else (Desktop, Downloads, ...)
- The folder is **self-contained**: move it, share it, open on another computer, ...

```
MyProject/  
├── data/  
│   └── tree_data.csv  
├── R/  
│   ├── clean_data.R  
│   └── statistics.R  
└── figures/
```

RStudio projects

An **RStudio project** turns this folder into a workspace for R:

- It's just a normal folder with an **.Rproj** file in it
- Opening the .Rproj file opens the entire project in RStudio
- Helps R to find your project files

```
MyProject/  
├── data/  
│   └── tree_data.csv  
├── R/  
│   ├── clean_data.R  
│   └── statistics.R  
├── figures/  
└── MyProject.Rproj
```

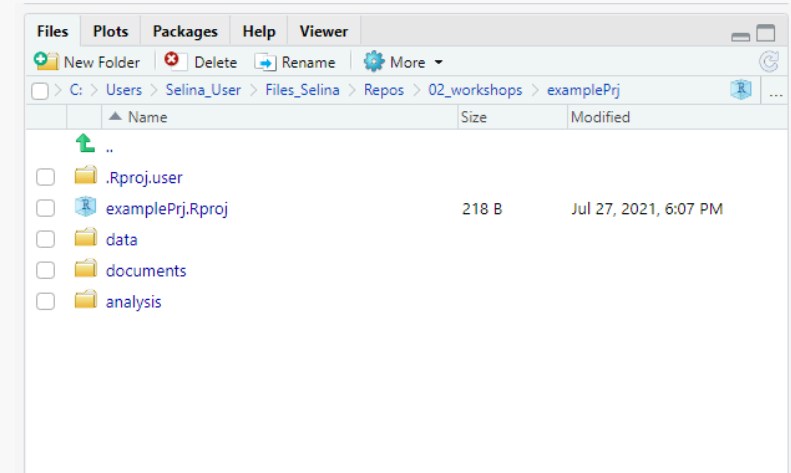
Same folder, plus an .Rproj file

Create an RStudio project

Create a project from scratch:

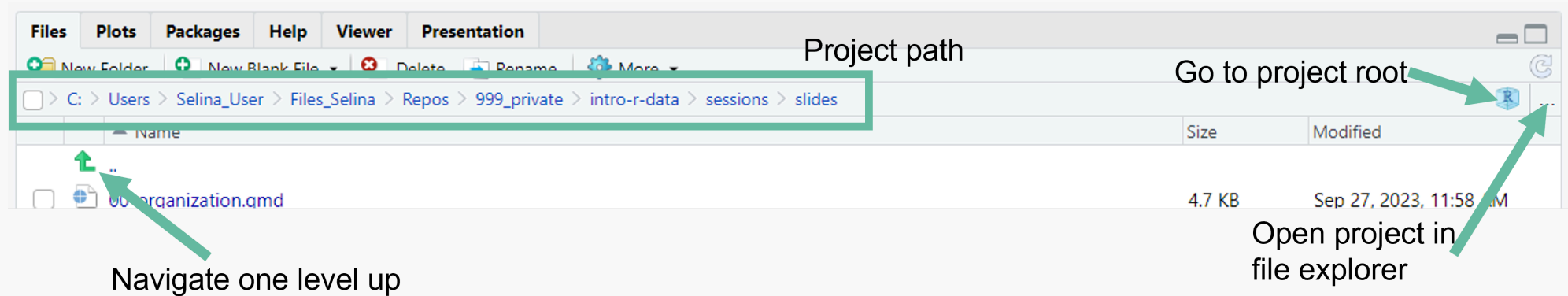
1. **File -> New Project -> New Directory -> New Project**
2. Enter a directory name (this will be the name of your project)
3. Choose the Directory where the project should be initiated
4. **Create Project**

RStudio will now create and open the project for you.



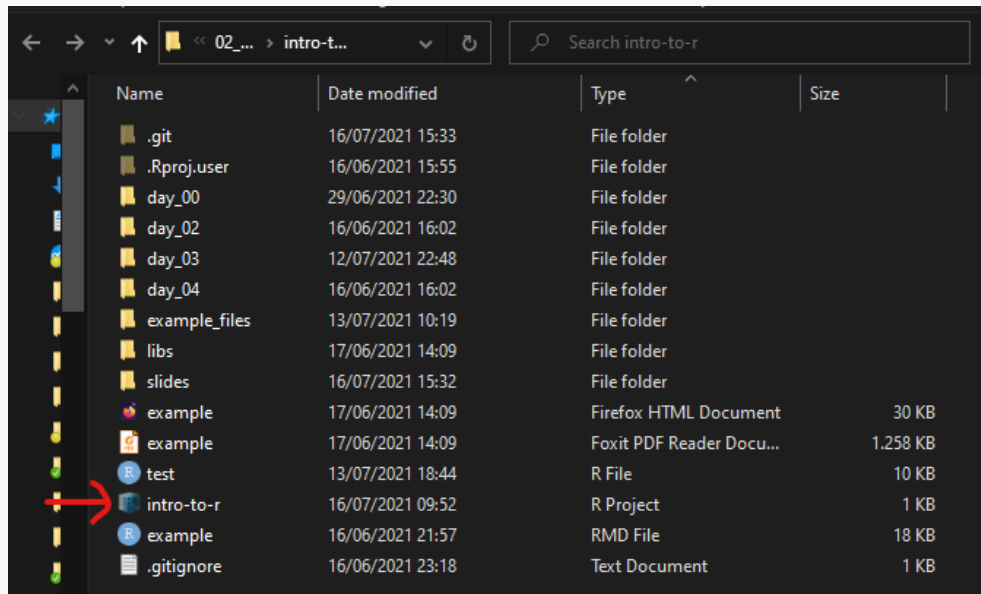
Example RStudio project structure in the Files pane

Navigate an RStudio project

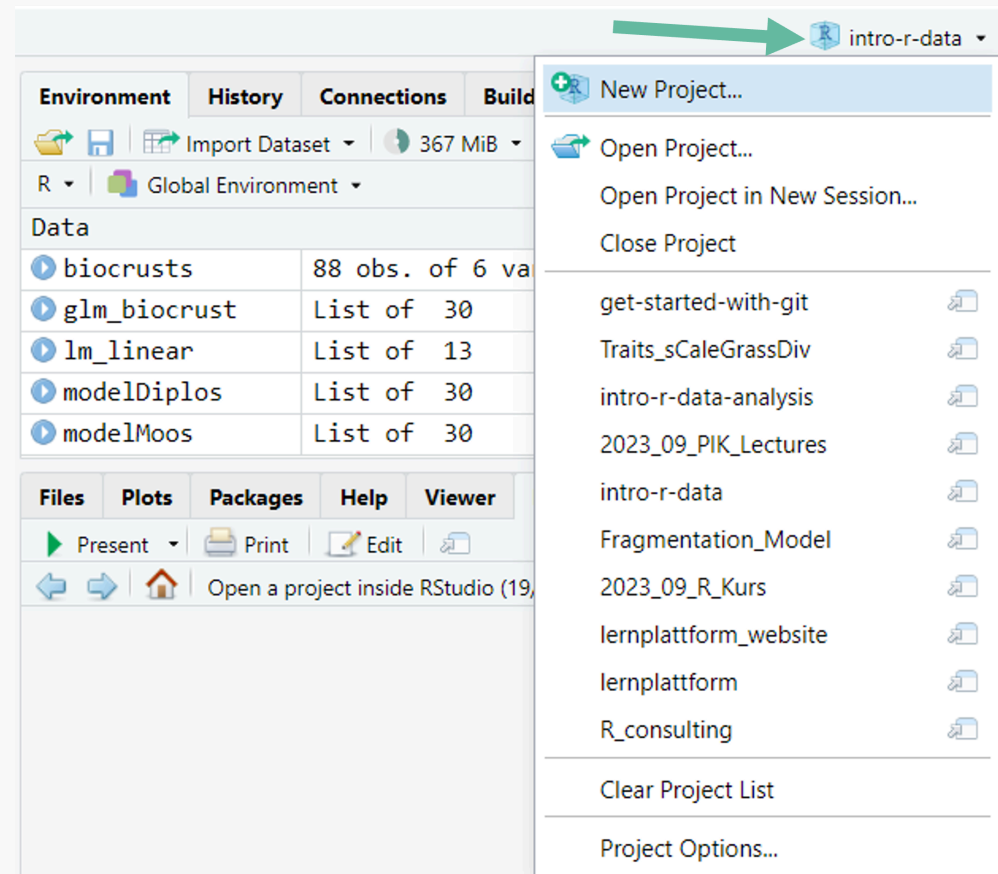


Open an RStudio project

From file explorer/finder:
Double click on the .Rproj file



From inside RStudio:
Click the project symbol on the top right



Now you

Task (15 min)

Set up your own RStudio project for this workshop

Find the task description [here](#)